ARCHITECTURE BLUEPRINT | PHASE 1

Industrial IoT Asset & Telemetry Platform

Core foundation: domain model, relational schema and verification

Repository	https://github.com/GoDZz412/STY-1004-Capstone-Project
Team	Gabriel (Member A) · Neel (Member B) · Billy (Member C)
Stack	.NET 10 · EF Core 10 · SQL Server · MSTest
Status	All Phase 1 code merged to **main** (cdc82c2): build 0 warnings, 209/209 unit tests passing
Date	6 November 2026

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1. System Overview

1.1 Problem statement

A mid-sized industrial site, for example a water-treatment plant or a mining and processing facility, runs hundreds of connected devices: pumps, motors, compressors, sensors and gateways. Today the device inventory lives in spreadsheets, sensor readings are not stored centrally, threshold alerts are configured inconsistently from device to device, and maintenance history is kept on paper. Simple questions are hard to answer: which devices exist, what state they are in, what they have been reading, which limits should raise an alarm, and who last worked on them.

The **Industrial IoT Asset & Telemetry Platform** provides one system of record for **assets**, their **telemetry**, the **alert rules** that watch it, and their **maintenance history**. Phase 1 builds the validated domain model and the relational database that all later phases rely on.

1.2 Stakeholders

Stakeholder	Interest	Platform capability
Plant operators	See which devices are active, under maintenance or retired; notice abnormal readings	Device status lifecycle; telemetry history; alert rules with severity
Maintenance technicians	Record the work done and look up a device's history	Immutable maintenance log per device
Reliability / site engineers	Define alert thresholds consistently across devices	Alert rules per device and metric (GreaterThan / LessThan; Info, Warning, Critical)
IT / OT administrators	A data store that can be rebuilt reliably	SQL Server schema from EF Core migrations; JSON seed data
Plant management	Trustworthy records and traceable history	Validated domain rules; maintenance history cannot be deleted silently

1.3 Platform scope

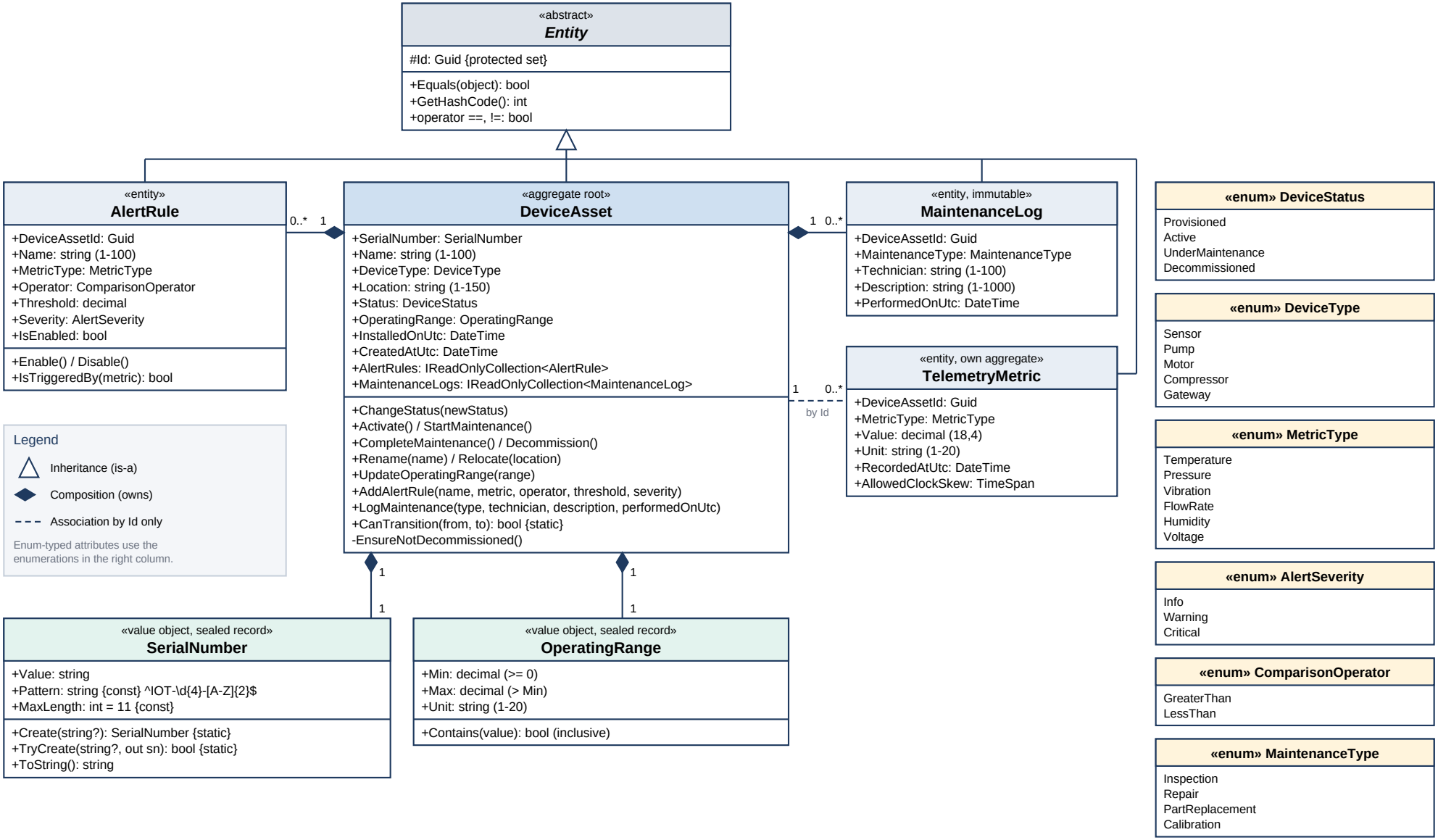
Delivered in Phase 1	Planned for later phases
Domain layer: 4 entities, 2 value objects, 6 enumerations, 6 domain exception types, all invariants enforced in constructors and methods Application layer: repository interfaces, DTOs with mappings, service contracts (interfaces only) Infrastructure: EF Core DbContext, Fluent API configurations, repositories, **InitialCreate** migration, JSON seed loader (8 devices, 10 alert rules, 50 telemetry readings, 8 maintenance logs) Verification: 209 unit tests (MSTest, SQLite in-memory for repository tests) This Architecture Blueprint	Service implementations and use cases APIs Telemetry ingestion Alert evaluation Dashboards

1.4 Solution architecture

Project	Responsibility	References
IoTPlatform.Domain	Pure business entities, value objects, enums and domain exceptions. Zero dependencies.	none
IoTPlatform.Application	Repository interfaces, DTOs and mappings, service contracts, IUnitOfWork.	Domain
IoTPlatform.Infrastructure	DbContext, Fluent API configurations, SQL Server repositories, migrations, JSON seed loader, DI registration.	Application, Domain
IoTPlatform.UnitTests	MSTest suite for domain rules, mappings and repository contracts (SQLite in-memory).	all three

Technology: .NET 10 and C#; Entity Framework Core 10.0.14 with SQL Server (LocalDB per team member, shared as code through configurations, migrations and seed JSON); MSTest 4.0.2 with EF Core Sqlite (in-memory) and coverlet for tests. Work is organised on GitHub with one branch per member and one pull request per task, each approved before merging into main.

2. Domain Model Diagram



*UML class diagram of **IoTPlatform.Domain**. DeviceAsset is the aggregate root and owns its AlertRules and MaintenanceLogs (composition, 1 to 0..*). TelemetryMetric is a separate, high-volume aggregate that references its device by DeviceAssetId only. SerialNumber and OperatingRange are immutable value objects.*

Domain model notes

Concept	Kind	Key rules
DeviceAsset	Aggregate root	Starts as Provisioned; Name 1-100, Location 1-150; install date not in the future; Rename / Relocate / UpdateOperatingRange / AddAlertRule / LogMaintenance refused once Decommissioned (DeviceDecommissionedException).
AlertRule	Child entity	Name 1-100; starts enabled; IsTriggeredBy is true only if the rule is enabled, same device, same metric type, and the value is strictly beyond the threshold (equal does not trigger).
MaintenanceLog	Child entity, immutable	Technician 1-100, Description 1-1000; performed date not in the future; no edit methods.
TelemetryMetric	Separate aggregate	Unit 1-20; timestamp may be at most 5 minutes in the future (clock skew); no collection on DeviceAsset.
SerialNumber	Value object	Mask IOT-####-AA (regex ^IOT-\d{4}-[A-Z]{2}$); trimmed and upper-cased before validation; InvalidSerialNumberException.
OperatingRange	Value object	Min >= 0, Max > Min, unit required (max 20); Contains is inclusive; InvalidOperatingRangeException.
Entity	Abstract base	Guid Id generated in the domain (Guid.Empty rejected); equality by Id and type.

Device status lifecycle

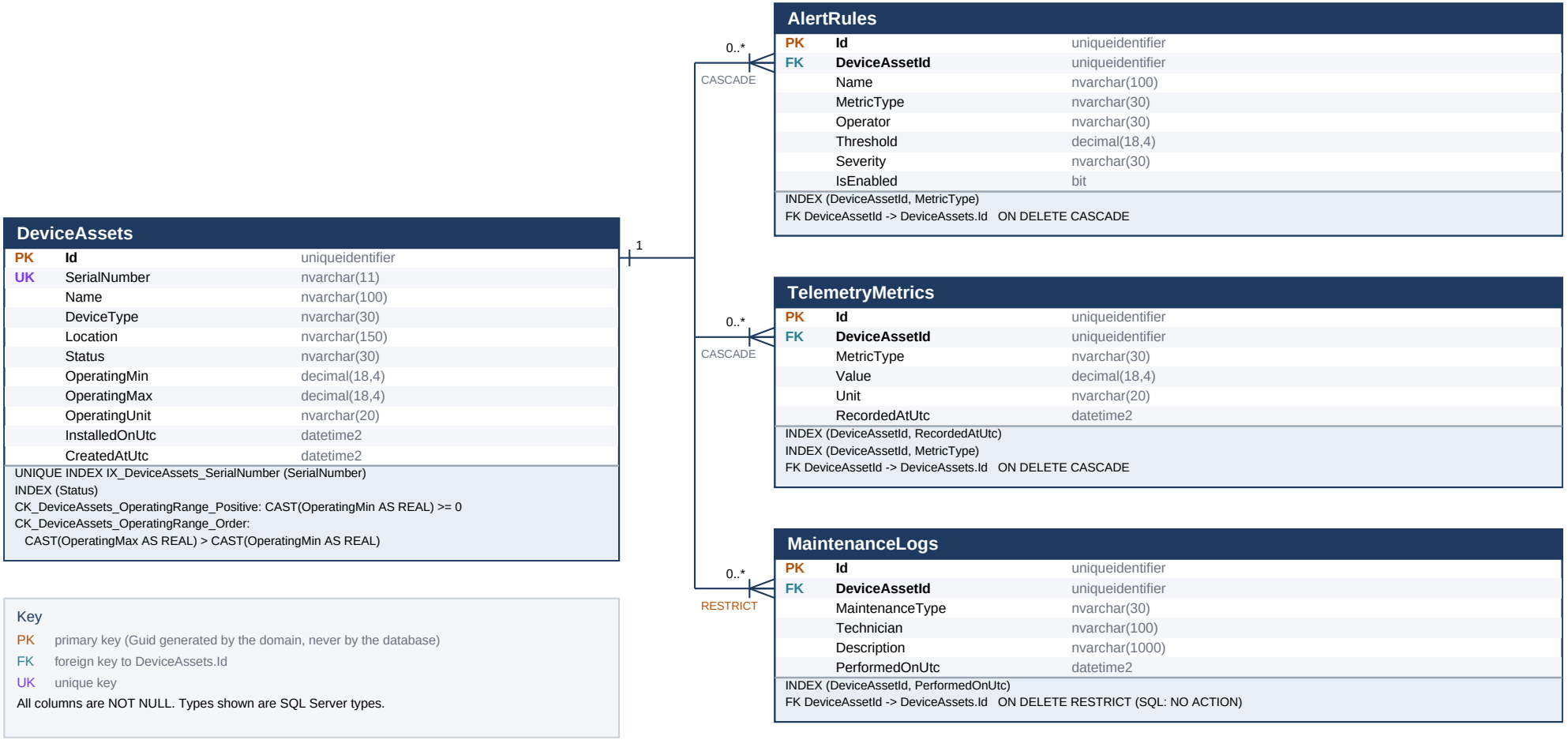
Allowed transitions are held in `DeviceAsset.AllowedTransitions` and tested for all 16 cells of the table. Moving to the same status is invalid, and Decommissioned is terminal. An invalid move throws `InvalidStatusTransitionException(from, to)`.

From \ To	Provisioned	Active	UnderMaintenance	Decommissioned
Provisioned	-	Allowed	-	Allowed
Active	-	-	Allowed	Allowed
UnderMaintenance	-	Allowed	-	Allowed
Decommissioned	-	-	-	-

Enumerations (stored as strings in the database)

Enum	Values
DeviceStatus	Provisioned, Active, UnderMaintenance, Decommissioned
DeviceType	Sensor, Pump, Motor, Compressor, Gateway
MetricType	Temperature, Pressure, Vibration, FlowRate, Humidity, Voltage
AlertSeverity	Info, Warning, Critical
ComparisonOperator	GreaterThan, LessThan
MaintenanceType	Inspection, Repair, PartReplacement, Calibration

3. Entity Relationship Diagram (ERD)



*Relational schema produced by the **InitialCreate** migration (SQL Server). One DeviceAsset has zero or more AlertRules, TelemetryMetrics and MaintenanceLogs. Deleting a device cascades to its rules and telemetry, but is blocked while it has maintenance history (decommission the device instead).*

Schema notes

Constraints per table

Every table has the primary key **Id** (uniqueidentifier).

Table	Foreign key / delete rule	Indexes	Check constraints
DeviceAssets	(parent table)	UNIQUE IX_DeviceAssets_SerialNumber (SerialNumber) (Status)	CK_DeviceAssets_OperatingRange_Positive: CAST(OperatingMin AS REAL) >= 0 CK_DeviceAssets_OperatingRange_Order: CAST(OperatingMax AS REAL) > CAST(OperatingMin AS REAL)
AlertRules	DeviceAssetId -> DeviceAssets.Id **CASCADE**	(DeviceAssetId, MetricType)	-
TelemetryMetrics	DeviceAssetId -> DeviceAssets.Id **CASCADE**	(DeviceAssetId, RecordedAtUtc) (DeviceAssetId, MetricType)	-
MaintenanceLogs	DeviceAssetId -> DeviceAssets.Id **RESTRICT** (SQL: NO ACTION)	(DeviceAssetId, PerformedOnUtc)	-

Type mapping conventions

Domain type	SQL Server type	Notes
Guid (Id, DeviceAssetId)	uniqueidentifier	Ids are generated by the domain (ValueGeneratedNever), never by the database
enum	nvarchar(30)	Stored as readable strings (HasConversion<string>)
decimal	decimal(18,4)	Threshold, Value, OperatingMin, OperatingMax
DateTime (UTC)	datetime2	All timestamps are UTC
bool	bit	AlertRules.IsEnabled (default true)
string	nvarchar(n)	Lengths reuse the domain constants (Name 100, Location 150, Technician 100, Description 1000, Unit 20)
SerialNumber (value object)	nvarchar(11)	Converted to and from string; unique index
OperatingRange (value object)	3 columns	Owned type flattened to OperatingMin, OperatingMax, OperatingUnit

The check constraints use CAST(... AS REAL) because the unit tests run on SQLite, which stores decimals as text. A unit test (`MigrationTests`) fails if a configuration changes without a matching new migration, and the generated SQL was compared line by line with this ERD.

4. Team Contributions

The entries below are taken from the contribution statements and handoff notes the members wrote themselves. Only Gabriel's first task (A1) is compiled from the project's wave handoff records. Commit hashes can be reproduced with `git log --oneline --author="<git name>"`.

4.1 Team summary

Member	Git identity	Role	Unit tests authored
Gabriel (Member A)	GitHub Gab07Con23; commits as Gabriel-C	Device Asset and domain foundation	113
Neel Patel (Member B)	GitHub neel7473; commits as Neel Sanjaybhai Patel	Telemetry and alerts	41
Billy Tran (Member C)	GitHub GoDZz412	Maintenance and persistence plumbing	55

Total: 209 unit tests, all passing on main.

4.2 Gabriel, Member A: Device Asset and domain foundation

Task	Contributions	Key commits
A1 Domain foundation (PR #6)	Entity base class, Guard, the six enums, the domain exceptions, SerialNumber and OperatingRange value objects (from the project handoff records).	`see PR #6`
A2 DeviceAsset aggregate (PR #18)	**DeviceAsset** aggregate root: constructor validated with `Guard`, starts as Provisioned; lifecycle with `AllowedTransitions`, `CanTransition`, `ChangeStatus` and the Activate / StartMaintenance / CompleteMaintenance / Decommission shortcuts; Rename, Relocate and UpdateOperatingRange blocked when decommissioned; read-only AlertRules and MaintenanceLogs collections, with marked spots left for B and C. **DeviceAssetTests** cover valid and invalid creation and the full 4x4 transition table.	**`e23b1c8`** **`fe741a2`** **`5a3b8f2`** **`6b87b1f`**

A3 Persistence + application layer (PR #22)	The **DeviceAssets** table, its repository and the application contracts: `IDeviceAssetRepository`, `IDeviceAssetService` (interface only) and `DeviceAssetDto` with `ToDto()`.
	DeviceAssetConfiguration: owned OperatingRange (OperatingMin, OperatingMax, OperatingUnit), SerialNumber value converter with unique index `IX_DeviceAssets_SerialNumber`, enums as strings with a Status index, CAST check constraints, ValueGeneratedNever.
	DeviceAssetRepository and the DeviceAssets DbSet. 17 tests: 1 mapping, 9 repository (SQLite), 7 Entity equality.
		`feat(application): add IDeviceAssetRepository, IDeviceAssetService and DeviceAssetDto`
		`feat(application): add DeviceAsset ToDto mapping`
		`test(application): add DeviceAssetMappingsTests for ToDto`
		`feat(infrastructure): add DeviceAssetConfiguration with owned range`
		`feat(infrastructure): add DeviceAssetRepository and DeviceAssets DbSet`
		`test(infrastructure): add DeviceAssetRepositoryTests on SQLite`
		`test(domain): add Entity equality tests`
		`hashes: 7d2b48b bc098db 890afe2 51260df 179284b 931e12f d0ad85d 502849a 3e810d2`

A4 Domain model diagram (PR #30)	`blueprint/domain-model.mmd`: Mermaid classDiagram of Entity, the four entities, the two value objects and the six enums, with inheritance, composition and the Id-only link to TelemetryMetric.	**`158bd88`** `docs(blueprint): add domain model class diagram`
A5 Seed data (PR #30)	`devices.json`: the 8 seed devices (pumps, motor, compressor, sensors, gateway) in every lifecycle status, UTF-8 without BOM.	**`0e390a7`** `feat(seeding): add devices.json seed data (8 devices)`

4.3 Neel Patel, Member B: Telemetry and alerts

Task	Contributions	Key commits
B1 Domain	Created the `TelemetryMetric` and `AlertRule` domain entities and their unit tests. Implemented and tested alert-rule enable/disable and trigger behavior, including metric, operator, threshold, device, and enabled-state cases.	`5d93086` `f94e94e`
B2 Application	Added `IAlertRuleRepository` and `ITelemetryMetricRepository` contracts, `AlertRuleDto` and `TelemetryMetricDto`, their `ToDto()` mappings, and the `ITelemetryService` interface.	`2bac8cd`
B2 domain method, delivered with B3	Implemented `DeviceAsset.AddAlertRule()` and its tests for a valid device and a decommissioned device.	`b3fdbe8`
B3 Infrastructure and persistence	Added `AlertRuleConfiguration` and `TelemetryMetricConfiguration`, the alert-rule and telemetry repository implementations, the two DbSet properties, and repository tests.	`b3fdbe8`
B4 Seed data	Created `alert-rules.json` with 10 rules and `telemetry.json` with 50 readings, linked by device serial number.	`822146e`
B4 Documentation	Prepared `blueprint/system-overview.md` for the Phase 1 Architecture Blueprint.	`123ed1a`

The earlier first-draft application use case (92dd2f0) was withdrawn during Billy's B2 corrections and is not listed as a completed deliverable.

4.4 Billy Tran, Member C: Maintenance and persistence plumbing

Task	Contributions	Key commits
Setup and coordination	Created the repository and the four-project solution and set up the work environment. Wrote the roadmap, the task briefs, the shared EF configuration and seed data contracts, and reviewed and approved every pull request, tidying formatting after merges.	**69b600d** `Work Environment Setup` **f72e33f** `style: tidy A3 leftovers (DbContext indentation, stray comments)` **006e012** `style: tidy B3 formatting (AddAlertRule indentation, usings, final newlines)`
C1 Persistence plumbing (PR #1)	**IoTPlatformDbContext** (provider-neutral, implements `IUnitOfWork`, applies every configuration automatically) and the `IUnitOfWork` contract. `DesignTimeDbContextFactory` for the dotnet ef tools (`IOTPLATFORM_CONNECTION` or LocalDB). **SqliteTestDatabase**: the shared SQLite in-memory helper used by every repository test.	**25ce244** `feat(infrastructure): add IoTPlatformDbContext, IUnitOfWork and SQLite test helper` **c31e0d1** `feat(infrastructure): add design-time DbContext factory for EF tooling`
C2 MaintenanceLog entity (PR #11)	**MaintenanceLog** entity, immutable and validated with `Guard`: device Id required, Technician 1-100, Description 1-1000, performed date not in the future. **MaintenanceLogTests** cover required fields, length limits and future dates. `PersistenceConstants.EnumMaxLength` (30) shared by every enum column.	**6a99f8d** `feat(domain): add MaintenanceLog entity with invariant tests` **6fd887e** `feat(infrastructure): add PersistenceConstants for enum column length`
C3 Maintenance persistence (PR #24)	`DeviceAsset.LogMaintenance()`, refused once decommissioned; `IMaintenanceLogRepository`, `MaintenanceLogDto` with `ToDto()`. **MaintenanceLogConfiguration**: FK Restrict (history is never deleted with its device), index (DeviceAssetId, PerformedOnUtc). **MaintenanceLogRepository** (newest first) and the MaintenanceLogs DbSet; coverlet collector for coverage reports. 8 tests: 2 domain, 1 mapping, 5 repository (SQLite).	**6ca2069** `feat(domain): add DeviceAsset.LogMaintenance and coverage collector` **ea6952b** `feat(application): add maintenance log repository contract and DTO` **b38e187** `feat(infrastructure): add maintenance log configuration and repository`
C4 Migration and DI (PR #33)	**InitialCreate** migration (4 tables, 3 foreign keys, 6 indexes, 2 check constraints), applied to LocalDB. **MigrationTests** guard: fails if a configuration changes without a new migration. **AddInfrastructure**: SQL Server DbContext, `IUnitOfWork` as the same scoped context, and the four repositories. 10 tests.	**a2991a4** `feat(infrastructure): add InitialCreate migration with pending-changes guard test` **472d2ec** `feat(infrastructure): add AddInfrastructure DI registration`
C5 JSON seed loader (PR #33)	**JsonSeedLoader**: builds every device, rule, log and reading through the domain constructors and methods; idempotent; one save; errors name the file and record. `maintenance-logs.json` (8 logs). `UseSeeding` so `dotnet ef database update` also loads the data. 14 tests, run on the team's real seed files.	**36d260f** `feat(seeding): add JSON seed loader with maintenance log seed data`
ERD	`blueprint/erd.mmd`: Mermaid ERD of the four tables with SQL types, keys, indexes and check constraints, cross-checked against the generated SQL and the InitialCreate migration.	**f53911a** `docs(blueprint): draft ERD for the Phase 1 schema` **1e40442** `docs(blueprint): match ERD delete rule to the generated SQL`